Cardiovascular System (CARD) · NPLEX-P1-CARD-11

CV-11 Hemodynamics and Blood Flow Regulation

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-11-01-C01				
NPLEX-P1-CARD-11-01-C02				
NPLEX-P1-CARD-11-01-C03				
NPLEX-P1-CARD-11-01-C04				
NPLEX-P1-CARD-11-01-C05				
NPLEX-P1-CARD-11-01-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-11-01-C01-BD1

Write the flow equations and Poiseuille's law from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C01 Activity: NPLEX-P1-CARD-11-01-C01-BD1 Date: _____

Without looking at your notes, write the equation that relates flow, pressure difference and resistance, then rearrange it for resistance and write why resistance is calculated rather than measured. Write the whole-body version of the same equation using mean arterial pressure, cardiac output and total peripheral resistance, and put in the resting values the notes gave for each one. Next write Poiseuille's law in full, name every symbol, and state for each variable whether resistance changes in direct or inverse proportion and by what power. Then write two or three sentences explaining why radius is raised to the fourth power, using the shape of the velocity profile in laminar flow. Finish by listing the three assumptions Poiseuille's law makes and what a living vessel does instead in each case.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-11-01-C01-BD2

Work out what single changes do to resistance and flow

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C01 Activity: NPLEX-P1-CARD-11-01-C01-BD2 Date: _____

Hold the pressure gradient constant and work out each of the following, writing the factor of change for resistance and for flow and one line of reasoning for each: the radius is halved, the radius is doubled, the radius widens by about 19%, viscosity doubles, and vessel length doubles. Then write a clinical example for each from the notes. Next, a man has a hematocrit of 68% (reference range about 40 to 45%). Write what happens to viscosity, to resistance, to flow and to oxygen delivery, and explain why oxygen delivery can fall even though the blood carries more oxygen per unit volume. Name three other changes in blood composition that raise viscosity. Finish by writing what rouleaux formation and a cell-poor layer at the wall each do to apparent viscosity and in which vessels each one matters.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-11-01-C01-PR01

Predict the effect of a focal arterial stenosis

A short segment of a muscular artery narrows so that its internal radius falls to half of normal, while the pressure gradient across the limb stays the same at first. Predict each item below, write the direction and size of the change where the notes let you give a number, and give the relationship or equation that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the resistance of the narrowed segment		
What happens to flow through that segment if the pressure gradient does not change		
What happens to the cross-sectional area of the lumen and to blood velocity inside the narrowed segment		
What happens to the Reynolds number in and just beyond the stenosis		
What is heard over the vessel and why		
What happens to that sound as the stenosis becomes critical and flow falls		
What happens to the resistance of the same stenosis if the diseased segment is long and diffuse instead of short		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-11-01-C01-Q01

Arteriolar dilation and flow

A 34-year-old woman begins moderate exercise. Arterioles in the quadriceps dilate so that their internal radius increases by approximately 19%, while mean arterial pressure is held constant by the baroreceptor reflex. Blood viscosity and vessel length do not change. Which of the following best describes the change in blood flow through those arterioles?

A. Flow increases by about 19%

B. Flow increases by about 50%

C. Flow approximately doubles

D. Flow increases approximately 16-fold

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-11-01-C02-BD1

Series and parallel resistance, and the pressure profile

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C02 Activity: NPLEX-P1-CARD-11-01-C02-BD1 Date: _____

Write the rule for adding resistances in series and the rule for adding them in parallel, each as an equation. Under the series rule write which quantity is the same in every segment and which segment takes the largest pressure drop, and explain why from the relationship between pressure drop, flow and resistance. Under the parallel rule write which quantity is the same in every branch, what happens to total resistance as branches are added, and what n identical branches of resistance R give. Work both rules once with numbers: three segments of 10, 20 and 20 units in series, and then the same three as parallel branches. Next build a table from memory with one row for the aorta and large arteries, one for small arteries and arterioles, one for capillaries and one for venules and veins, and columns for mean pressure, share of total resistance, total cross-sectional area and share of blood volume. Finish by writing the three or four features of arterioles that make them hold the largest adjustable share of resistance, and why billions of capillaries in parallel do not.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-11-01-C02-BD2

Adding and removing parallel paths, and beds placed in series

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C02 Activity: NPLEX-P1-CARD-11-01-C02-BD2 Date: _____

A surgeon creates a large arteriovenous fistula in the forearm. Write what the fistula does to total peripheral resistance and why, then write in order what happens to mean arterial pressure, what the baroreceptors do, what happens to cardiac output and what happens to diastolic pressure and pulse pressure. Write what a very large fistula does to the heart over time. Next, a patient undergoes a mid-thigh amputation. Write what happens to total peripheral resistance and explain it from the parallel rule. Then write the six-step sequence from a large bed dilating to flow being redirected while mean arterial pressure is defended, one step per line. Finish with the capillary beds that lie in series rather than parallel: name three from the notes, write what raised downstream resistance does to upstream capillary pressure, and use that to explain how portal hypertension develops.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-11-01-C03-BD1

Velocity against total cross-sectional area

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C03 Activity: NPLEX-P1-CARD-11-01-C03-BD1 Date: _____

Write the equation that links flow, velocity and cross-sectional area, then write why velocity at each level of the systemic circuit varies inversely with the total cross-sectional area at that level. Draw one set of axes with vessel segments from aorta to vena cava along the horizontal axis, and sketch total cross-sectional area and mean velocity as two curves on it. Mark the aortic and capillary velocities with the values from the notes and write the approximate ratio of capillary area to aortic area. Write why capillary velocity has to be lowest and how long a red cell spends in a systemic capillary, then write what that transit time is for. Finish by writing why velocity rises again in the veins but stays below the aortic value.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-11-01-C03-BD2

Turbulence at the bedside

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C03 Activity: NPLEX-P1-CARD-11-01-C03-BD2 Date: _____

Write the Reynolds number as an equation, name each variable, and state which direction each one moves the number. Write which of those variables barely changes in the body and which three are the clinical ones. Then explain in two or three sentences why turbulent flow is audible and why it needs a larger pressure gradient for the same flow. Next, a woman has a hemoglobin of 7.2 g/dL (reference range about 12 to 16 g/dL) and a soft systolic murmur across structurally normal valves. Rebuild the sequence from the low hemoglobin to the murmur, one step per line, and write why anemia is the classic cause of a flow murmur while polycythemia tends to abolish one. Then write what makes each of these audible or silent and why: a carotid stenosis, a critical carotid stenosis, an arteriovenous fistula, and the brachial artery under a blood pressure cuff at a pressure just below systolic and then below diastolic. Finish by writing where in the arterial tree shear stress is low and oscillating, what that does to endothelial nitric oxide, and what tends to form there.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-11-01-C03-RT01

Rebuild the sequence from anemia to a flow murmur

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C03 Activity: NPLEX-P1-CARD-11-01-C03-RT01 Date: _____

Rebuild the seven-step sequence that runs from a falling hemoglobin and hematocrit to a systolic flow murmur across normal valves, one step per line and in order. Beside the viscosity step write what viscosity does to resistance and to the Reynolds number, and note that these two effects pull the murmur in opposite directions through different variables. Beside the arteriolar step write which stimulus dilates the arterioles. Beside the cardiac output step write which component of cardiac output rises most. Then write the three other states from the notes that produce a flow murmur by the same reasoning.

Rebuild from memory: From anemia to a flow murmur

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-11-01-C04-BD1

Compliance, the arterial reservoir and pulse pressure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C04 Activity: NPLEX-P1-CARD-11-01-C04-BD1 Date: _____

Write the equation for compliance, name its reciprocal, and write how much more compliant systemic veins are than the matching arteries and what that explains about where blood sits. Then describe the reservoir function of the elastic arteries in your own words: what the wall does during ejection, what it does during diastole, and the three things that function achieves. Write the definition of pulse pressure with its normal value, then write what pulse pressure is roughly proportional to. Write what governs how fast stored volume runs off during diastole, give the time constant as a product of two variables, and list the three conditions that let diastolic pressure fall further. Finish with the formula that estimates mean arterial pressure from diastolic and pulse pressure, and write at which heart rates that estimate becomes less accurate and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-11-01-C04-BD2

Predict the pressure pattern in a stiff aorta

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C04 Activity: NPLEX-P1-CARD-11-01-C04-BD2 Date: _____

Two people have the same stroke volume of 70 mL, the same heart rate and the same total peripheral resistance. One is 25 years old with a pressure of 120/80 mm Hg. The other is 80 years old with aortic compliance halved. Work out the second person's pulse pressure, systolic pressure, diastolic pressure and mean arterial pressure, show your reasoning for each, and write why mean arterial pressure barely moves while systolic and diastolic pressures both change. Then write the wall changes that lower aortic compliance with age, name the three diseases that speed them up, and explain what faster pulse wave velocity does to the timing of the reflected wave and to the systolic peak. Write the two downstream consequences for the left ventricle and for coronary perfusion, and say which part of the ventricular wall is most exposed. Finish by sorting each of the following into widens or narrows pulse pressure, with the mechanism in one line: aortic regurgitation, hemorrhage, aortic stenosis, hyperthyroidism, cardiac tamponade, arteriovenous fistula, exercise.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-11-01-C05-BD1

Draw the capillary and its Starling forces

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C05 Activity: NPLEX-P1-CARD-11-01-C05-BD1 Date: _____

Draw a capillary with its arterial end on the left and its venous end on the right, and add a lymphatic vessel in the interstitium beside it. Write the Starling equation for net filtration in full and name every term, including the filtration coefficient and the reflection coefficient. On your drawing mark the capillary hydrostatic pressure at each end, the interstitial hydrostatic pressure, the plasma oncotic pressure and the interstitial oncotic pressure, using the teaching values from the notes, and draw an arrow for each force in the direction it moves fluid. Calculate the net pressure at the arterial end and at the venous end, show the working, and say which way fluid moves at each end. Write how much fluid the lymphatics return in a day and what else they carry back. Finish by writing which protein supplies most of the plasma oncotic pressure and roughly what share, and where the reflection coefficient and the filtration coefficient are unusually high or low.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-11-01-C05-BD2

Work out the edema in four patients

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C05 Activity: NPLEX-P1-CARD-11-01-C05-BD2 Date: _____

For each patient below, write which Starling term changes, which direction it moves, whether the fluid is a transudate or an exudate, whether the swelling pits, and where in the body the swelling appears first and why. Patient 1 has nephrotic syndrome with a plasma albumin of 1.8 g/dL (reference range about 3.5 to 5.0 g/dL). Patient 2 has right heart failure with raised jugular venous pressure. Patient 3 has a deep burn to one forearm. Patient 4 had axillary nodes removed and radiated two years ago. For patient 1, take plasma oncotic pressure from 28 mm Hg down to 16 mm Hg and recalculate the net pressure at both ends of the capillary, showing the working, then write why the edema is generalized. For patient 4, write why the swelling stops pitting as the years pass. Finish by writing the three safety factors that delay edema, how much the notes say capillary hydrostatic pressure can rise before edema becomes obvious, and what changes in the interstitium once its pressure passes about 0 mm Hg.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-11-01-C05-CP01

The four mechanisms of edema side by side

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Raised capillary hydrostatic pressure	Low plasma oncotic pressure	Increased capillary permeability	Lymphatic obstruction
Which Starling term changes and in which direction				
Two or three causes from the notes				
Transudate or exudate				
Protein content of the fluid and specific gravity				
Does the swelling pit, and does that change over time				
Generalized or localized, and where it appears first				

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-11-01-C05-CA01

Abdominal swelling and ankle edema in a man with liver disease

A 57-year-old man comes to the clinic because of six weeks of increasing abdominal girth and swelling of both ankles. He has a ten-year history of heavy alcohol use. Temperature is 36.8 degrees C (98.2 degrees F), pulse is 92/min, respirations are 16/min, and blood pressure is 104/62 mm Hg. The abdomen is distended with shifting dullness, and there are dilated veins over the anterior abdominal wall. Pressing over each medial malleolus leaves an indentation that takes several seconds to fill. The jugular venous pressure is not raised and the lungs are clear. Serum albumin is 2.1 g/dL (reference range 3.5 to 5.0 g/dL), total protein is 5.4 g/dL, and the fluid removed at paracentesis has a specific gravity of 1.009.

1. Recognize. Name the two Starling terms that are abnormal in this patient and say which vascular bed each one acts on.

2. Explain. Explain how portal hypertension develops in this patient, using the arrangement of the gut and splenic capillaries relative to the hepatic sinusoids, and explain how the low albumin acts on the same capillaries.

3. Differentiate. Write how the Starling picture here differs from the edema of right heart failure and from the edema of lymphatic obstruction, naming the findings in this case that separate them.

4. Investigate. Write which measurements and physical findings you would use to confirm the mechanism, and what value or appearance each one would have.

5. Decide. State which single mechanism explains why the fluid collects in the abdomen first rather than only in the legs, and state what the specific gravity of the fluid tells you about the capillary wall.

6. Justify. Justify your answer from the Starling equation, naming the term that each finding reflects, and explain why the safety factors that normally delay edema are overwhelmed in this patient.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-11-01-C06-BD1

The four layers of control over arteriolar tone

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C06 Activity: NPLEX-P1-CARD-11-01-C06-BD1 Date: _____

Start with the smooth muscle itself: write the sequence from a rise in cytosolic calcium to a contracted arteriole, naming the calcium binding protein and the two enzymes that set the level of myosin light chain phosphorylation, then write the two general routes by which a dilator relaxes the same cell. Now organise the four layers of control on one page, with a heading for each. Under metabolic control, list the substances that accumulate and dilate, name the channel many of them open, and define active hyperemia and reactive hyperemia. Under myogenic control, write the sequence from stretch to constriction naming the two channel types involved. Under endothelial control, list the stimuli that activate the nitric oxide pathway, name the other dilator and the constrictor the endothelium releases, and write what acetylcholine does when the endothelium is damaged and why. Under neural and hormonal control, write which receptor and G protein sympathetic norepinephrine uses, what the main neural dilator mechanism is and why, and then list the hormones that constrict and those that dilate with the receptor and second messenger for each. Finish by writing the approximate mean arterial pressure range over which brain and kidney autoregulate, and what chronic hypertension does to that range.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-11-01-C06-BD2

Which control wins in which bed

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C06 Activity: NPLEX-P1-CARD-11-01-C06-BD2 Date: _____

Build a table from memory with one row each for the coronary, cerebral, skeletal muscle, skin, pulmonary and renal beds, and columns for dominant control, response to hypoxia and the effect of sympathetic stimulation. Then answer each of the following in two or three sentences, using the mechanism rather than the table entry. Why does the heart have to raise flow rather than extract more oxygen when demand rises, and what does that do to the alpha-1 effect on coronary arterioles. Why does a patient hyperventilating during a panic attack feel lightheaded, and which molecule crosses the blood-brain barrier to produce that effect while another one does not. Why does flow to exercising muscle rise about 20-fold despite high sympathetic outflow, and what is that override called. Why is the skin pale and cool after a hemorrhage while coronary and cerebral flow are preserved. Why do pulmonary vessels constrict to low alveolar oxygen when every systemic bed dilates, what does that achieve locally, and what happens when the hypoxia involves the whole lung. Finish with a patient whose blood pressure has run at 170/100 mm Hg for years and is lowered quickly to 110/70 mm Hg: write why that person can become hypoperfused at a pressure a normotensive person tolerates.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-11-01-C06-RT01

Rebuild the nitric oxide pathway

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-11 Hemodynamics and Blood Flow Regulation
Competency: NPLEX-P1-CARD-11-01-C06 Activity: NPLEX-P1-CARD-11-01-C06-RT01 Date: _____

Rebuild the seven-step nitric oxide pathway from the stimulus at the endothelium to the dilated arteriole, one step per line and in order, and draw it as two cells side by side with the endothelial cell on the left and the smooth muscle cell on the right so that each step sits in the cell where it happens. Name the enzyme in the endothelium, its substrate and both of its products. Name the enzyme in the smooth muscle that nitric oxide activates, the second messenger it makes and the kinase that messenger activates. At the last two steps write what happens to cytosolic calcium and to myosin light chain phosphorylation. Then write where a nitrate drug enters this pathway and why that matters when the endothelium is damaged.

Rebuild from memory: Nitric oxide pathway from endothelium to smooth muscle

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-11-01-C06-Q02

Regional flow during hyperventilation

A 22-year-old woman is brought to the emergency department during a panic attack. Respirations are 34/min, pulse is 108/min, and blood pressure is 128/76 mm Hg. She reports lightheadedness and tingling in both hands. Arterial blood gas analysis shows a PaCO2 of 22 mm Hg and a PaO2 of 108 mm Hg. Which of the following best explains her lightheadedness?

A. A rise in perivascular pH that constricts cerebral arterioles and reduces cerebral blood flow

B. A fall in mean arterial pressure below the lower limit of cerebral autoregulation

C. Inhibition of oxygen-sensitive potassium channels in cerebral arteriolar smooth muscle

D. Loss of myogenic tone in cerebral arterioles with a rise in cerebral blood flow

Why did you pick it?